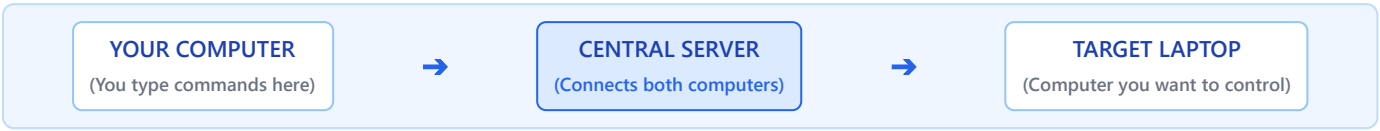
LaptopCtl — Easy Step-by-Step User Guide

Control Any Windows Laptop Remotely with Simple Code Commands

CONCEPT How LaptopCtl Works in Plain English



Target laptops **never need router port forwarding or special Wi-Fi settings**. Both computers talk to your central server through an encrypted tunnel.

QUICKSTART Test It on Your PC in 3 Minutes (Zero Cloud Needed)

You can test the entire system right now on your computer with **3 PowerShell windows**:

1 WINDOW 1 — Start the Backend Server

Open **PowerShell** and paste this exact command:

```
# Navigate to Backend folder and start the server
cd "c:\Users\bhara\Downloads\laptopctl\src\LaptopCtl.Backend"
dotnet run
```

Leave this window open! It will say: `Now listening on: http://localhost:5000`.

2 WINDOW 2 — Create Admin Account & Open Web Dashboard

Open a **second PowerShell window** and create your administrator login:

```
# Create an admin account (12+ character password)
$body = @{ email = "admin@laptopctl.corp"; password = "MasterPassword123!" } | ConvertTo-Json
Invoke-RestMethod -Uri "http://localhost:5000/api/auth/register" -Method Post -Body $body -ContentType
"application/json"
```

Visual Dashboard is Ready! Open your browser and go to: `http://localhost:5000`
You will see the live visual dashboard with specs, device list, and interactive terminal!

3 WINDOW 3 — Log in with the CLI Remote Control

Open a **third PowerShell window** to use the `laptopctl` command-line tool:

```
cd "c:\Users\bhara\Downloads\laptopctl\src\LaptopCtl.Controller"
dotnet run -- login -b "http://localhost:5000" --email "admin@laptopctl.corp" --password "MasterPasswo
rd123!"
```

It will say: `[✓] Logged in successfully as admin@laptopctl.corp`. Your computer is now ready!

STEP-BY-STEP How to Connect (Enroll) Any Windows Laptop

To control any laptop (in the other room or across the world), you only need to run **one code command** on that laptop once.

1 Generate an Enrollment Code from your Controller

In your controller window, ask the server for a new temporary code:

```
dotnet run -- enroll-code -b "http://localhost:5000"
```

The screen will print a box with an 8-character code, like: `ABC12345`.

2 Run the 1-Line Command on the Target Laptop

Walk over to the laptop you want to control (or tell your friend):

1. Press **Windows Key**, type **PowerShell**, right-click it and choose **"Run as Administrator"**.
2. Copy and paste this single command and press Enter:

```
irm "http://YOUR_SERVER_IP:5000/install.ps1" | iex; Install-LaptopCtl -EnrollmentCode "ABC12345" -Back
end "http://YOUR_SERVER_IP:5000"
```

3 What Happens Automatically in 10 Seconds

Action	What the computer does behind the scenes
1. Security Lock	Creates `C:\ProgramData\LaptopCtl` and locks it so normal users cannot touch it.
2. Encryption	Generates a unique RSA military-grade encryption key using Windows DPAPI.
3. Registration	Contacts your server, enters code `ABC12345`, and receives a private device token.
4. Background Service	Registers Windows Service `LaptopCtlAgent` that starts automatically on every boot.

Done! The laptop is now connected. Even if the laptop restarts or switches Wi-Fi networks, it will automatically reconnect to your server silently!

VISUAL UI Using the Modern Web Dashboard

If you prefer clicking buttons instead of typing commands, open `http://localhost:5000` in your browser:

Dashboard Tab	What you can do with one click
Enrolled Devices	See all your laptops with green ● `Online` or gray ○ `Offline` status lights and specs.
Diagnostics & Specs	View CPU model, RAM bars, hard drive free space gauges, and running background processes.
Remote Terminal	A black terminal screen in your browser where you can type commands and run them remotely.
File Explorer	Browse files and folders on the remote laptop and click "Download".
Enroll Computer	1-Click button to generate a new enrollment code with a copyable PowerShell snippet.

REMOTE CONTROL Complete Command Cheat Sheet

Run these from your controller folder (`src\LaptopCtl.Controller`):

What you want to do	Exact Code Command to Run
See all your laptops	`dotnet run -- devices`
Inspect Laptop Specs (CPU, RAM, OS)	`dotnet run -- info <device-id>`
See What Programs Are Running	`dotnet run -- processes <device-id>`
Check Hard Drive Free Space	`dotnet run -- disk <device-id>`
Check Wi-Fi / IP Addresses	`dotnet run -- network <device-id>`
Run ANY Command on That Laptop	`dotnet run -- exec <device-id> "ipconfig"` `dotnet run -- exec <device-id> "Get-Service"`
Look at Files on the Remote Laptop	`dotnet run -- files list <device-id> "C:\Users\Public"`
Send a File to the Remote Laptop	`dotnet run -- files upload <device-id> "local.zip" "C:\Users\Public\local.zip"`
Download a File from Remote Laptop	`dotnet run -- files download <device-id> "C:\Users\Public\file.txt" "./file.txt"`
Remove / Disconnect a Laptop	`dotnet run -- uninstall <device-id>`

HOSTING How to Host Online for Free (Control from Anywhere)

To control laptops anywhere in the world when you are not at home, host the server online:

Option A: 1-Click Free Hosting on Railway or Render

1. Create a free account at **railway.app** or **render.com**.
2. Click **New Project → Deploy from GitHub** (or select Dockerfile).
3. Point it to the `deploy/Dockerfile.backend` in this project.
4. Add a free PostgreSQL database plugin.
5. Your server is now live at a permanent address like `https://my-laptopctl.up.railway.app`!

Option B: On a $5 Cloud Server (DigitalOcean / AWS / Linode / Hetzner)

On any Linux VPS server with Docker installed, run this single command inside the `deploy` folder:

```
docker compose up -d
```

Everything (Postgres, Web Relay, Nginx, Dashboard) starts automatically on ports 80/443!

UNINSTALL How to Remove from a Laptop

If you ever want to completely delete the agent from a laptop, open PowerShell as Administrator and run:

```
& "c:\Users\bhara\Downloads\laptopctl\installer\uninstall.ps1"
```

This stops and removes the background service and wipes all files cleanly.